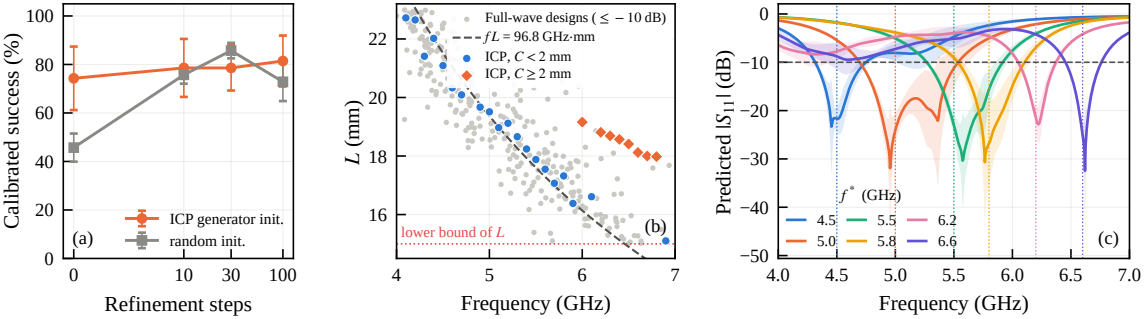

Calibrated success (%)
100
80
60
40
20
0
(a)
ICP generator init.
random init.
0
10
30
100
Refinement steps
L (mm)
22
20
18
16
Full-wave designs (≤ − 10 dB)
fL = 96.8 GHz·mm
ICP, C < 2 mm
ICP, C ≥ 2 mm
lower bound of L
(b)
4
5
6
7
Frequency (GHz)
Predicted |S11| (dB)
0
−10
−20
−30
−40
−50
f* (GHz)
4.5
5.0
5.5
5.8
6.2
6.6
(c)
4.0
4.5
5.0
5.5
6.0
6.5
7.0
Frequency (GHz)